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Effect of Different Teaching Methods on Test Scores

Introduction:

There is an extensive amount of research that has been done on different teaching methods in order to determine which one is the most effective (“The Flipped Classroom”). This is an important topic for teachers and students alike. I chose to center my research around two teaching methods in particular, the standard method, and the flipped method. The standard teaching method involves lectures or other forms of in-class teaching of new material. The teacher assigns homework – where students have an opportunity to apply their learning – outside of the classroom. The flipped teaching method is where students are assigned online lectures or readings to do at home, and practice applying their knowledge through hands-on work when they are in the classroom (Frontiers in Psychology). Initial research suggests that flipped learning may have a positive effect on students’ performance in class. Research also suggests that the flipped teaching method may be more effective for some class subjects than for others (“The Flipped Classroom”). I am thus focusing on the following research questions: Which teaching method helps students perform the best? And, is any teaching method more effective for a particular age group of students?

Methods:

I chose to use a qualitative by qualitative treatment structure, with randomized complete block design for the experiment. The two treatments are teaching method and age of students. Each treatment has two levels, teaching method has flipped (f) and traditional (t), and type of class has either high school-age (h) students or college-age (c) students. This makes 4 possible treatment combinations: $f \times h$, $f \times c$, $t \times h$, and $t \times c$. The blocks were done by the class subject. I decided to do this because I think that some teaching methods may produce better results in particular class subjects. The response variable was the mean test scores of the class. When doing initial research, test scores seemed to be something that could be effectively represented using a Beta distribution because they can be expressed as a continuous proportion from 0 to 1 (“The Flipped Classroom”). Through a power analysis we determined that 6 blocks should be used because this resulted in 80% power for the method $\times$ type (teaching method by type/age of class) interaction term. In the experiment itself, we used 6 blocks and the 4 different treatment combinations were represented in each block, which resulted in 24 total observations. The methods used in this experiment are displayed well by the What Would Fisher Do (WWFD) table ([Figure 1](#)). This table shows 23 total degrees of freedom (df), method $\times$ type with 1 df, and the

residual term with 15 df. In our analysis we will check to make sure that the degrees of freedom line up with our WWFD table (Stroup).

Model:

The response variable for my experiment – test scores – is distributed by a Beta distribution: $Y \sim \text{Beta}(\mu_{ijk}, \phi)$, where μ is the mean and ϕ is the shape parameter. The linear predictor for our model is: $\eta_{ijk} = \eta + \alpha_i + \beta_j + (\alpha\beta)_{ij} + b_k$ and the link function looks like: $\eta_{ijk} = \log(\mu_{ijk}/1-\mu_{ijk})$. In the model, η is the overall intercept and α_i represents the effect of the i^{th} level of teaching method. β_j is effect of the type of class at the j^{th} level, $(\alpha\beta)_{ij}$ is the interaction term between teaching method and type of class, and b_k is the blocking term. A condition of the model is that our blocks are independent and normally distributed, with mean zero and a variance of sigma-squared: $b_k \sim \text{NI}(0, \sigma^2_B)$.

Analysis:

When conducting the analysis, the following process is carried out. First, we will look at the covariance parameter estimate for blocking to see if blocking by class subject was effective. Then we will look at the interaction plot and the Type III Tests of Fixed Effects table to see if the interaction term is significant. If interaction is present then we will look at simple effects, otherwise look at main effects. Then we will want to look at overall means, and odds ratios to determine confidence intervals.

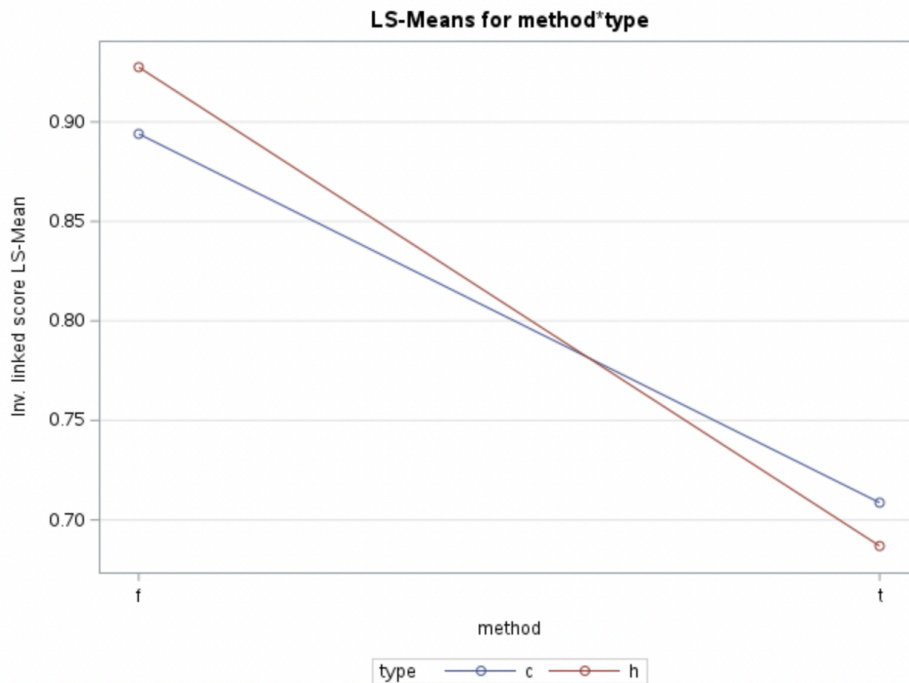
Results:

Data was generated for the experiment and it was printed in a table to make sure everything was inputted correctly ([2-Input Data](#)). The Covariance Parameter Estimates table looked like this:

Covariance Parameter Estimates			
Cov Parm	Subject	Estimate	Standard Error
Intercept	block	0.004516	.
Scale		302.81	101.31

[Figure 3](#)

This table shows a 0.004516 estimate for the effect of blocking. Because this estimate is close to zero it means that blocking by class subject was not able to capture much extra variance that was not already captured by the model. However, because this estimate is not exactly zero, we cannot say that blocking was totally ineffective. The next step in our analysis is to look at the interaction plot for method by type. The plot is shown below:



[Figure 4](#)

From this plot we can see that there is potentially an interaction between method and type because the size of the vertical difference at both levels look significant. We can also make some initial observations about the treatment means. It looks as if the flipped method results in higher mean test scores for both college and high school-age students. The mean test scores for the traditional method at both age groups appear to be quite a bit lower. We can next move to Type III tests to confirm our initial inspection of the interaction plot and to see if the method x type interaction is significant:

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
method	1	15	474.50	<.0001
type	1	15	5.19	0.0378
method*type	1	15	14.22	0.0018

Figure 5

We can first see that our numerator degrees of freedom is 1 for method x type, and the denominator degrees of freedom is 15. This lines up with our WWFD table (Figure 1). The next important thing we notice is that the method by type term yields a p-value of 0.0018. This is significant at an $\alpha=0.05$ (95% significance level). This tells us that we will be focusing on simple effects in our further analysis.

The next important step in our analysis is to look at the overall means for each combination of treatments:

method*type Least Squares Means													
method	type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper	Mean	Standard Error Mean	Lower Mean	Upper Mean
f	c	2.1321	0.08059	15	26.46	<.0001	0.05	1.9604	2.3039	0.8940	0.007637	0.8766	0.9092
f	h	2.5493	0.09417	15	27.07	<.0001	0.05	2.3486	2.7500	0.9275	0.006331	0.9128	0.9399
t	c	0.8888	0.05842	15	15.21	<.0001	0.05	0.7643	1.0134	0.7087	0.01206	0.6823	0.7337
t	h	0.7856	0.05746	15	13.67	<.0001	0.05	0.6631	0.9080	0.6869	0.01236	0.6600	0.7126

Figure 6

The only important column we want to examine in this table is the means column. We can see that the flipped teaching method with high school students yields the highest mean of 0.9275. Second is the flipped method with college students which has a mean of 0.8940. Far below these two combinations, were the traditional methods for college and high school, yielding 0.7087 and 0.6869 for their respective means.

Now looking at simple effects:

Simple Effect Comparisons of method*type Least Squares Means By method													
Simple Effect Level	type	_type	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper	Odds Ratio	Lower Odds Ratio	Upper Odds Ratio
method f	c	h	-0.4171	0.1175	15	-3.55	0.0029	0.05	-0.6676	-0.1667	0.659	0.513	0.846
method t	c	h	0.1033	0.07217	15	1.43	0.1729	0.05	-0.05054	0.2571	1.109	0.951	1.293

Figure 7

At the simple effect level for method f, the odds ratio comparing college and high school is 0.659. We will flip this, so our new odds ratio is equal to $1/0.659$ which is 1.5. We can then flip our lower and upper odds ratios to get a confidence interval (CI) that runs from approximately 1.2 to 1.9. Because 1 is not included in this new confidence interval, we can say that high school is significantly better than college when the flipped teaching method is employed. This lines up with a significant p-value of 0.0029.

For method t, the odds ratio is 1.109. The confidence interval is 0.951 to 1.293. This interval includes 1 which means that the difference in the means between college and high school when using the traditional method is insignificant. This matches up with a p-value of 0.1729 which is not significant at an $\alpha=0.05$.

Looking at more simple effects:

Simple Effect Comparisons of method*type Least Squares Means By type													
Simple Effect Level	method	_method	Estimate	Standard Error	DF	t Value	Pr > t	Alpha	Lower	Upper	Odds Ratio	Lower Odds Ratio	Upper Odds Ratio
type c	f	t	1.2433	0.09164	15	13.57	<.0001	0.05	1.0480	1.4386	3.467	2.852	4.215
type h	f	t	1.7637	0.1032	15	17.09	<.0001	0.05	1.5437	1.9837	5.834	4.682	7.270

[Figure 8](#)

When comparing the flipped and traditional methods for college age students, we get an odds ratio of 3.467. This means that the flipped teaching method is approximately 3.5 times more likely to yield a higher test score than the traditional method. The confidence interval goes from approximately 2.9 to 4.2 and does not include 1 which means that the flipped method produces significantly higher mean test scores compared to the traditional method for college-age students.

For the simple effect level of high school, we observe an odds ratio of 5.834 and a confidence interval of 4.7 to 7.3. This CI does not include 1, which means the flipped method results in significantly higher mean test scores than the traditional at the high school level.

Conclusion:

It is clear that the flipped teaching method produces higher mean test scores than the traditional method. When the flipped method is used, it is significantly better for high school age students than for college age students. When the traditional method is used, there is no significant difference between the mean test scores for both age groups. Overall, keeping in mind that this is an experiment that is using simulated data, the recommendation would be to use the flipped classroom over the traditional for both high school and college-age students.

Appendix

Figure 1 (WWFD Table):

Design	df	Treatment	df	Combined	df
Class	$6-1=5$			Class	$6-1=5$
		method	$2-1=1$	method	$2-1=1$
		type	$2-1=1$	type	$2-1=1$
		method*type	$1*1=1$	method*type	$1*1=1$
e.u.(class)	$(4-1)*6=18$	“parallels”	20	e.u.(class) (m,t,m*t)	$18-1-1-1=15$
Total	$(6*4)-1=23$	Total	$24-1=23$	Total	$24-1=23$

2 - Input Data:

```

data scores;
input block method $ type $ score;
datalines;
1 t h 0.65
2 t h 0.63
3 t h 0.72
4 t h 0.71
5 t h 0.70
6 t h 0.71
1 f h 0.94
2 f h 0.93
3 f h 0.93
4 f h 0.92
5 f h 0.92
6 f h 0.93
1 t c 0.75
2 t c 0.68
3 t c 0.72
4 t c 0.73
5 t c 0.66
6 t c 0.71
1 f c 0.87
2 f c 0.86
3 f c 0.92
4 f c 0.90
5 f c 0.92
6 f c 0.89
;
proc print data=scores;
run;

```

Figure 3 (Cov. parm. block, deleting some excess code) :

```
proc glimmix data=scores method=quadrature;
  class block method type;
  model score=method|type/d=beta;
  random intercept / subject=block;
run;
```

Figure 4 (Interaction Plot):

```
proc glimmix data=scores method=quadrature;
  class block method type;
  model score=method|type/d=beta;
  random intercept / subject=block;
  lsmeans method*type/plot=meanplot(sliceby=type ilink join);
run;
```

Figure 5 (Type III Tests – same as Figure 3 code):

```
proc glimmix data=scores method=quadrature;
  class block method type;
  model score=method|type/d=beta;
  random intercept / subject=block;
run;
```

Figures 6, 7, and 8 (LS Means and Simple Effects):

```
proc glimmix data=scores method=quadrature;
  class block method type;
  model score=method|type/d=beta;
  random intercept / subject=block;
  lsmeans method*type/ ilink cl oddsratio slicediff=(method type);
run;
```

References:

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